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Interface bus combining

Abstract

Circuits and methods enabling common control of an agent device by two or more buses, particularly MIPI RFFE serial buses. In essence, the invention provides flagging signals designating completed register write operations to denote which of two registers are active, such that synchronization is accomplished in a clock-free manner. One embodiment includes at least two decoders, each including a common register and a bus (S/P) decoder coupled to a respective bus and to the common register. The S/P decoder asserts a write-complete signal when a write operation to a corresponding common register is completed. A multiplexer has at least two selectable input bus ports coupled to the common registers within the at least two decoders. A selection circuit selects an input bus port of the multiplexer in response to the assertion of a last write-complete signal from the S/P decoders.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS—CLAIM OF PRIORITY (1) The present application is a continuation application of co-pending and commonly assigned U.S. application Ser. No. 17/354,530, filed Jun. 22, 2021, for a “Interface Bus Combining”, to issue on Jan. 30, 2024 as U.S. Pat. No. 11,886,228, which is herein incorporated by reference in its entirety.

BACKGROUND

(1) Technical Field

(1) This invention relates to electronic circuits, and more particularly to electronic integrated circuits interconnected by a communications bus.

(2) Background

(2) Modern electronic circuits, particularly radio frequency (RF) electronic circuits, are commonly implemented by interconnecting one or more integrated circuits (“ICs”, also known as “dies”), each providing one or more desired functions, such as amplification, modulation/demodulation, tuning, switching, etc. It is also common to embed at least one IC and additional external circuit elements (e.g., filters, tuning elements, etc.) in a circuit module configured to be coupled to other circuit modules and/or additional external circuit elements or system elements (e.g., user controls, antennas, etc.).

(3) In the realm of RF electronics, RF communication systems typically include “RF front-end” (RFFE) circuitry, which is a generic term for all of the circuitry between a radio antenna up to and including the mixer stage of a radio. An industry standard serial bus has been developed by the Mobile Industry Processor Interface (MIPI) Alliance to interconnect sets of circuit modules for RFFE circuitry. In particular, the MIPI RF Front-End (RFFE) Control Interface serial bus has been widely adopted in a variety of RF systems, particularly mobile wireless systems.

(4) The MIPI RFFE serial bus is a serial, two-wire, controller/agent interface originally designed for controlling a variety of RF front end devices, such as amplifiers, antenna switches, filters, etc. (the MIPI bus has since been extended for controlling other modules or devices as well). One wire is a bi-directional serial data line (SDATA), and the other wire is a “Bus Controller” generated synchronous clock (SCLK). A third wire, VIO, may be used as a voltage reference/supply, to control power consumption, and to provide reset and enable functions for the MIPI RFFE serial bus.

(5) FIG. 1 is a block diagram showing an example of a portion of an RF Front-End circuit **100**. A MIPI controller **102** is coupled by a serial data line SDATA and clock line SCLK to three agent circuits, a power amplifier circuit **104**, a filter circuit **106**, and a one-to-many switch circuit **108** (to avoid clutter, connections to VIO are omitted). Each of the coupled agent circuits **104**, **106**, **108** includes an internal Decoder that converts serial data on SDATA into register addresses and 8-bit command/data words which may be stored in addressed registers.

(6) Taking a particular implementation of the MIPI RFFE serial bus as one example, the maximum number of agent devices on the serial bus is 15. The SCLK signal line provides a clock signal that is always driven by the controller device on the serial bus, at a specified maximum clock rate (e.g., 26 MHz MIPI RFFE 1.0, and 52 MHz for MIPI RFFE 2.0). There is always a controller device, and up to 15 agent devices, each coupled to the SCLK and SDATA signal lines through a respective control and status (C/S) decoder interface. In general, agent devices may support several operating modes (e.g., “startup”, “active state”, “shutdown”, “low-power mode”, etc.) and device-specific functions (e.g., operating parameters for amplifiers, switches, programmable tuning and/or filter components, etc.), which are selected by issuing command messages through the bit-serial SDATA signal line under the control of the SCLK signal line.

(7) The SDATA signal line is driven high by the controller while the SCLK signal line remains low to initiate a transfer; this is called the Sequence Start Condition or SSC. The SDATA signal line is driven high for one clock period and then low for one clock period while the SCLK signal line is low. After an SSC event occurs, the SCLK and SDATA signal lines are driven by the controller device (e.g., MIPI controller **102**) to transfer a 12-bit command frame (plus 1-bit odd parity).

(8) The 12-bit command frame consists of two parts, a 4-bit agent Device Address and an 8-bit operational command. The first 4 bits of the command frame are an agent address, SA[3:0], corresponding to a unique agent identifier (USID). If SA[3:0] are all zeroes, then the command is broadcast to and acted upon by all agent devices. Commands that all devices must respond to may include, for example, commands to “go to low power mode” or “reset to a known state”. The next 8 bits of the command frame, C[8:0], are the operational command, comprising a 3-bit command type and a 5-bit Register Address, A[4:0]. An agent device having a USID that matches SA[3:0] decodes the operational command to determine the type of command and a designated Register Address within the agent device.

(9) FIG. 2 is a block diagram of a generic agent device **200**. The agent device **200** includes a decoder **202** which in turn includes a bus decoder **204**, which is a serial-to-parallel decoder for a MIPI RFFE serial bus. The bus decoder **204** has the characteristic that it receives and decodes WRITE commands, data, and an address of a register. Decoded commands and data from the bus decoder **204** are stored in a register file **206** that includes a set (e.g., 16 or 32) of addressable registers **208**. The commands and data are provided to controller circuitry **210**, in known manner.

(10) In some implementations of the MIPI RFFE serial bus, only two command types are supported: Register Read (for reading status information from an agent device) and Register Write (for providing control information to an agent device). The command types may be respectively indicated by a code in the first 3 bits of the 8-bit operational command (e.g., Read=binary “011”, Write=binary “010”). The last 5 bits of the 8-bit operational command designate a Register Address, A[4:0], within the agent device; accordingly, each agent device can support up to 32 registers using the Register Write and Register Read commands (in some embodiments of agent devices, only a subset of the addressable register space is supported).

(11) If a command indicates a Register Write command type, then the command frame is followed by a Write Data frame driven by the controller device. A single Write Data frame may contain from a few bits of payload data up to multiple (e.g., 16) bytes of payload data. Multiple Write Data frames may be transmitted in sequence to an addressed agent device. An addressed agent device will store received payload data in one or more addressed registers, and respond as appropriate to the function of the agent device.

(12) Other serial bus standards (e.g., the Serial Peripheral Interface, I2C, I2S, SMBus, and/or PMBus) typically have comparable command frame structures and command types. In addition, parallel bus standards (e.g., AMBA) are used in some systems.

(13) While the MIPI RFFE serial bus and similar buses generally provide adequate communication functionality for many RF Front-End circuits, a drawback is that multiple buses cannot be accommodated. For example, new cell phone models require simultaneous support of 4G and 5G technologies. Because of the differences in frequency bands and data rates between 4G and 5G technologies, separate transceivers and separate command buses (i.e., MIPI RFFE serial buses) may be used.

(14) For example, FIG. 3 is a block diagram **300** showing an example of a portion of a non-functional circuit having dual MIPI RFFE serial buses and dual RF Front-End circuits. A first MIPI controller **302a** is coupled by a serial data line SDATA and clock line SCLK to three agent circuits, a power amplifier circuit **304a**, a filter circuit **306b**, and a one-to-many switch circuit **308**. The three agent circuits coupled to the first MIPI controller **302a** may be, for example, to process a mid-band of frequencies. The one-to-many switch circuit **308** may be coupled to multiple antennas, and used to direct amplified and filtered mid-band frequencies to a selected one of those antennas.

(15) A desired second MIPI controller **302b** (shown in dashed outline) is coupled by a serial data line SDATA and clock line SCLK to three agent circuits, a power amplifier circuit **304b**, a filter circuit **306b**, and the one-to-many switch circuit **308**. (To avoid clutter, connections to VIO are omitted). The three agent circuits coupled to the second MIPI controller **302b** may be intended, for example, to process a high-band of frequencies. The one-to-many switch circuit **308** may be used to direct amplified and filtered high-band frequencies to a selected one of antennas. Thus, both the mid-band and high-band share the same one-to-many switch circuit **308**, and accordingly there would be a need to provide commands to the one-to-many switch circuit **308** from both the first and second MIPI controllers **302a**, **302b**.

(16) Unfortunately, the architecture shown in FIG. 3 is not possible using two MIPI RFFE buses because there are two clock domains with non-deterministic clocks. Thus, either bus could try to write commands to the internal register file of an agent device without regard for the activity of the other bus.

(17) A general method for receiving signals from two different clock domains is to use a synchronizer from one (secondary) clock domain to a primary clock domain. Then an arbiter (e.g., a mutual-exclusion or mutex circuit) may be used in the primary domain. However, this method requires that both clock signals from the two different clock domains be available at all times to the synchronizer. This cannot happen with a MIPI RFFE bus because clock signals are sent only when there is a transaction; accordingly, both clock signals from the two different clock domains would be always available for a synchronizer.

(18) Accordingly, there is a need for circuits and methods that enable common control of one or more agent devices by two or more buses, particularly MIPI RFFE serial buses. The present invention addresses this need.

SUMMARY

(19) The present invention encompasses circuits and methods that enable common control of one or more agent devices by two or more buses, particularly MIPI RFFE serial buses. In essence, the invention provides flagging signals designating completed register write operations to denote which of two registers are active, such that synchronization is accomplished in a clock-free manner.

(20) One embodiment comprises a circuit that enables common control of one or more agent devices by two or more buses, the circuit including: A circuit that enables common control of one or more agent devices by two or more buses, the circuit including: at least two decoders, each including at least one common register and a bus decoder coupled to a respective bus and to the at least one common register, the bus decoder configured to assert a write-complete signal when a write operation to a corresponding one of the at least one common register is completed; a

multiplexer having at least two selectable input bus ports each coupled to a respective one of the at least one common register within the at least two decoders; and a selection circuit, coupled to two or more of the bus decoders and to the multiplexer, the selection circuit configured to select an input bus port of the multiplexer in response to the assertion of a last write-complete signal from the coupled bus decoders.

(21) A method encompassed by the invention includes enabling common control of one or more agent devices by two or more buses by asserting a write-complete signal when a write operation from one of the two or more buses to a common register is completed, and selecting the content of the common register associated with one of the two or more buses in response to the assertion of a last write-complete signal.

(22) The details of one or more embodiments of the invention are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the invention will be apparent from the description and drawings, and from the claims.

Description

DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a block diagram showing an example of a portion of an RF Front-End circuit.

(2) FIG. 2 is a block diagram of a generic agent device.

(3) FIG. 3 is a block diagram showing an example of a portion of a non-functional circuit having dual MIPI RFFE serial buses and dual RF Front-End circuits.

(4) FIG. 4A is a block diagram showing an example of an RF Front-End agent device supporting control by dual MIPI RFFE serial buses.

(5) FIG. 4B is a block diagram of a circuit that accommodates common registers within four decoders corresponding to four MIPI controllers like those shown in FIG. 4A.

(6) FIG. 5 is a top plan view of a substrate that may be, for example, a printed circuit board or chip module substrate (e.g., a thin-film tile).

(7) FIG. 6 illustrates an exemplary prior art wireless communication environment comprising different wireless communication systems, and which may include one or more mobile wireless devices.

(8) FIG. 7 is a block diagram of a transceiver **700** that might be used in a wireless device, such as a cellular telephone, and which may beneficially incorporate an embodiment of the present invention for improved performance.

(9) FIG. 8 is a process flow chart showing one method for enabling common control of one or more agent devices by two or more buses.

(10) Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

(11) The present invention encompasses circuits and methods that enable common control of one or more agent devices by two or more buses, particularly MIPI RFFE serial buses. In essence, the invention provides flagging signals designating completed register write operations to denote which of two registers are active, such that synchronization is accomplished in a clock-free manner.

First Embodiment

(12) FIG. 4A is a block diagram showing an example of an RF Front-End agent device **400** supporting control by dual MIPI RFFE serial buses. Each agent device that requires common control by multiple buses is provided with a corresponding number of decoders. In the illustrated example, two MIPI controllers **401a**, **401b** are shown respectively coupled to corresponding decoders **402a**, **402b** by corresponding SCLK and SDATA serial lines.

(13) Each decoder **402a**, **402b** includes a bus decoder **404a**, **404b**, such as a serial-to-parallel decoder. Note that while the disclosed example embodiments show serial-to-parallel decoders, the

invention is applicable to embodiments using parallel busses in which a clock signal is associated with parallel address and data busses. Decoded commands from the bus decoders **404a**, **404b** are stored in a respective register file **406a**, **406b** that includes a set (e.g., 16 or 32) of addressable registers **408a**, **408b**. Ultimately, one or more of the addressable registers **408a**, **408b** will be coupled to controlled circuits; in the illustrated example, two controlled circuits **410**, **412** are shown. The controlled circuits **410**, **412** may include, for example, amplifiers, couplers, switches, programmable tuning, and/or filter components.

(14) In the illustrated example, the addressable registers **408a**, **408b** that will store commands and data that are not common to the controlled circuits **410**, **412** are uniquely labeled 0x01 . . . 0x07 in register file **406a** and 0x07 . . . 0x0E in register file **406b** (collectively, "0x0 #"). In the illustrated example, the addressable registers **408a**, **408b** that will store commands and data that are common to the controlled circuits **410**, **412** are labeled CMN0 and CMN1 (while the common registers are shown separated from the associated unique 0x0 # registers, all of the common and unique registers are part of a respective register file **406a** or **406b**).

(15) In the illustrated example, the 0x0 # registers of the decoders **402a**, **402b** may be coupled to one or more controlled circuits **410**, **412**. For example, controlled circuit **410** is shown as coupled by word-wide buses to the A set of 0x0 # registers from decoder **402a** and the B set of 0x0 # registers from decoder **402b**. Internal to controlled circuit **410** would be the necessary logic to utilize commands from either MIPI controller **401a**, **401b**.

(16) For controlled circuit **412**, the A set of 0x0 # registers from decoder **402a** and/or the B set of 0x0 # registers from decoder **402b** may be optionally coupled, is indicated by the dotted connection line. However, controlled circuit **412** may be of a type that responds only to commands from the common registers CMN0, CMN1, and accordingly may not be connected to the 0x0 # registers from decoder **402a** or **402b**.

(17) The bus decoders **404a**, **404b** may be essentially similar to conventional bus decoders, but with added output signals to indicate when certain Register Write operations are complete (i.e., all bits of a received command and/or data frame have been received over the corresponding SDATA serial line and stored in an addressed register). In the illustrated example, when a Register Write operation to the CMN0 register of a decoder **402a**, **402b** is complete, then a "Write Complete" output signal WC0 is momentarily asserted by the associated bus decoder **404a**, **404b**. Similarly, when a Register Write operation to the CMN1 register of a decoder **402a**, **402b** is complete, then a "Write Complete" output signal WC1 is momentarily asserted by the associated bus decoder **404a**, **404b**.

(18) Both of the controlled circuits **410**, **412** are shown as having a common input bus port InC, which is shown connected to an output bus port D of a respective multiplexer **414a**, **414b**. In the illustrated example, each multiplexer **414a**, **414b** has selectable input bus ports S.sub.1, S.sub.2, either of which may be coupled through to a corresponding output bus port D based on a selection control signal applied to a selector port C. In the illustrated example, for multiplexer **414a**, S.sub.1 is coupled to the CMN1 register of decoder **402a**, and S.sub.2 is coupled to the CMN1 register of decoder **402b**. For multiplexer **414b**, S.sub.1 is coupled to the CMN0 register of decoder **402a**, and S.sub.2 is coupled to the CMN0 register of decoder **402b**.

(19) The selector ports C of the multiplexers **414a**, **414b** are coupled to an output port Q of respective set-reset (S-R) flipflops **416a**, **416b**. In the illustrated example, for flipflop **416a**, input S is coupled to the WC0 signal from bus decoder **404a**, and input R is coupled to the WC0 signal from bus decoder **404a**. Similarly, for flipflop **416b**, input S is coupled to the WC1 signal from bus decoder **404a**, and input R is coupled to the WC1 signal from bus decoder **404a**.

(20) The circuit configuration in FIG. 4A allows either MIPI controller **401a**, **401b** to write to the common registers CMN0 and/or CMN1 of a respective decoder **402a**, **402b** and have that register content supplied to one or more controlled circuits **410**, **412** within an RF Front-End agent device **400**. A "last write wins" (LWW) policy is utilized to select which of the common registers CMN0

and/or CMN1 are coupled to the internal controlled circuits **410**, **412**. Basically, for any common register that exists in both decoders **402a**, **402b**, the LWW mechanism ensures that whichever register was written to last by its corresponding MIPI controller **401a**, **401b** is declared the winner. This policy allows for mutually exclusive arbitration between MIPI RFFE buses, and forms a deterministic rule for which of the two buses' data is effectively applied to the internal controlled circuits **410**, **412**. At the same time, the LWW policy provides design flexibility, in that a customer can select which bus to use for writing common commands and/or data to an RF Front-End agent device **400**.

(21) As an example of operation, suppose that MIPI controller **401a** writes a command to decoder **402a** addressed to register CMN0. Register CMN0 of decoder **402a** is coupled to the S.sub.1 input of multiplexer **314b**, while register CMN0 of decoder **402b** is coupled to the S.sub.2 input of multiplexer **314b**. When a Register Write operation to the CMN0 register of decoder **402a** is complete, then a Write Complete output signal WC0 is asserted by the associated bus decoder **404a**. That WC0 signal sets flipflop **416b**, which causes multiplexer **414b** to pass the contents of register CMN0 of decoder **402a** through to the controlled circuit **412**.

(22) Conversely, suppose that MIPI controller **401b** writes a command to decoder **402b** addressed to register CMN0. When a Register Write operation to the CMN0 register of decoder **402b** is complete, then a Write Complete output signal WC0 is asserted by the associated bus decoder **404b**. That WC0 signal resets flipflop **416b**, which causes multiplexer **414b** to pass the contents of register CMN0 of decoder **402b** through to the controlled circuit **412**.

(23) Whichever MIPI controller **401a**, **401b** last writes to its associated CMN0 register, the corresponding Write Complete output signal WC0 will cause the content of the last written CMN0 register to be coupled through to the controlled circuit **412**.

(24) In a similar fashion, suppose that MIPI controller **401a** writes a command to decoder **402a** addressed to register CMN1. Register CMN1 of decoder **402a** is coupled to the S.sub.1 input of multiplexer **314a**, while register CMN1 of decoder **402b** is coupled to the S.sub.2 input of multiplexer **314a**. When a Register Write operation to the CMN1 register of decoder **402a** is complete, then a Write Complete output signal WC1 is asserted by the associated bus decoder **404a**. That WC1 signal sets flipflop **416a**, which causes multiplexer **414a** to pass the contents of register CMN1 of decoder **402a** through to the controlled circuit **410**.

(25) Conversely, suppose that MIPI controller **401b** writes a command to decoder **402b** addressed to register CMN1. When a Register Write operation to the CMN1 register of decoder **402b** is complete, then a Write Complete output signal WC1 is asserted by the associated bus decoder **404b**. That WC1 signal resets flipflop **416a**, which causes multiplexer **414a** to pass the contents of register CMN1 of decoder **402b** through to the controlled circuit **410**.

(26) Again, whichever MIPI controller **401a**, **401b** last writes to its associated CMN1 register, the corresponding Write Complete output signal WC1 will cause the content of the last written CMN1 register to be coupled through to the controlled circuit **410**.

(27) There may be cases where the Write Complete output signals (e.g., WC0 and WC1) conflict. For example, in some 5G and 4G radio systems, MIPI commands may come from different integrated circuits without coordination. In the case where two or more bus decoders (e.g., **404a** and **404b**) concurrently output a Write Complete signal, there may be some contention as to which signal wins under the "last write wins" policy. Such contention may be resolved in a number of ways. For example, the S-R flipflops (e.g., **416a**, **416b**) may be designed to prefer the S (Set) input over the R (Reset) input if there is contention.

(28) The disclosed circuits and methods for directing control circuitry by multiple different interfaces provide a number of benefits, such as a deterministic policy (since the last write to either bus interface persists), convenience for a customer (since the customer may write to either bus interface), and ease of implementation (since the LWW policy is implemented without clocked circuits, avoiding considerable issues with clock synchronization and asynchronous clocking

domains). Implementation is simple, requiring little in the way of additions to a conventional MIPI RFFE architecture, and each MIPI controller remains independent.

Alternative Embodiments

(29) The concepts shown in FIG. 4A may be extended to additional MIPI controllers. For example, FIG. 4B is a block diagram of a circuit that accommodates common registers within four decoders corresponding to four MIPI controllers like those shown in FIG. 4A. The flipflops **416a**, **416b** of FIG. 4A would be replaced by a “radio button” circuit **420** that includes 4 flipflops **422a** . . . **422d**. Each flipflop **422a** . . . **422d** would be set by a corresponding Write Complete output signal **WC0x** from an associated bus decoder within a corresponding decoder (not shown, but like the bus decoders **404a**, **404b** and decoders **402a**, **402b** of FIG. 4A). In addition, each Write Complete output signal **WC0x** is coupled to the reset input of each other flipflop through an OR gate **424a** . . . **424d** associated with each other flipflop.

(30) Thus, for example, when Write Complete output signal **WC0.sub.0** is asserted, flipflop **422a** is set, and flipflops **422b** . . . **422d** are reset, and when Write Complete output signal **WC0.sub.3** is asserted, flipflop **422d** is set, and flipflops **422a** . . . **422c** are reset. The result is that the flipflops **422a** . . . **422d** and OR gates **424a** . . . **424d** function like a radio button circuit, in which the last Write Complete output signal **WC0x** to a flipflop is the only asserted output (e.g., a “1”) of the radio button circuit, and all other flipflops have the opposite logic state (e.g., “0”).

(31) The outputs of the flipflops **422a** . . . **422d** are coupled to a conventional priority encoder **426** which converts the input signals to a binary encoded output. The binary encoded output is applied to a multiplexer **428** to select one of four possible registers **CMN0.sub.0** . . . **CMN0.sub.3** to output to a controlled circuit (e.g., like controlled circuits **410**, **412** in FIG. 4A). The multiplexer **428** thus replaces one of the multiplexers **414a**, **414b** shown in FIG. 4A. If the multiplexer **428** is configured to respond to 4 direct selector inputs (e.g., **C0** . . . **C3**), rather than binary encoded selector bits, the priority encoder **426** may be omitted and the outputs of the flipflops **422a** . . . **422d** may be directly coupled to that type of multiplexer **428**.

(32) As should be appreciated, other circuits may be devised that provide “radio button” functionality for outputting the last written one of *m* inputs. Thus, the invention encompasses a selection circuit, coupled to two or more bus decoders on separate buses and to a multiplexer having input bus ports coupled to two or more common registers, the selection circuit being configured to select an input bus port of the multiplexer in response to the assertion of a last write-complete signal from the coupled bus decoders.

(33) The circuit of FIG. 4B may be replicated as may be desired for additional common registers **CMN #** and corresponding Write Complete output signals **WC0 #**. As should be clear, the “radio button” circuit **420** can be adapted to accommodate more than, or fewer than, four MIPI controllers (the case of two MIPI controllers essentially devolving down to the flipflop and multiplexer configuration shown in FIG. 4A).

(34) The concepts shown in FIGS. 4A and 4B may be extended to additional common registers **CMN #**. For example, the circuit of FIG. 4B may be replicated for an additional common register **CMN1** having corresponding Write Complete output signals **WC1**. The circuit of FIG. 4B may also be used with multiple common registers **CMN #** and multiple MIPI controllers.

(35) While the examples above have associated a Write Complete output signal **WC #** with a single common register **CMN #**, in some embodiments, one Write Complete output signal **WC #** may be associated with a set of two or more registers, such that when any register within the set is written to by a corresponding MIPI controller, the entire set of registers from the last-written MIPI bus is declared the winner. For example, referring to FIG. 4A, common registers **CMN0**, **CMN1** may be considered to be a set, such that the last write to either register within a decoder **402a**, **402b** from either MIPI control **401a**, **401b** will cause the contents of both registers **CMN0**, **CMN1** to be passed through multiplexer **414b** to controlled circuit **412** (of course, multiplexer **414b** must be coupled to both registers **CMN0**, **CMN1**).

(36) In some embodiments, a Write Complete output signal WC # may be associated with a field within a single common register CMN #. A field is a subset of bits within the register (e.g., an 8-bit register may have two 4-bit fields). Accordingly, for common registers that support masked writes (a known MIPI capability), a Write Complete output signal WC # may be asserted when the bits of a field are written, allowing the last written field to be coupled to a controlled circuit.

(37) Under certain conditions, a non-latching “last write” combinatorial circuit may be used rather than the S-R flipflops shown in FIG. 4A. For example, referring to FIG. 4A, the flipflops **416a**, **416b** may be replaced by OR gates if the common registers CMN # that are not going to be selected are first cleared (e.g., by a MIPI command), in which case the Write Complete output signal WC # for the common register CMN # that is going to be selected will cause the corresponding multiplexer to output that common register CMN #. Similarly, referring to FIG. 4B, the radio button circuit **420** may be omitted if the common registers CMN # that are not going to be selected are first cleared. Further, if the multiplexer **428** is configured to respond to direct selector inputs, the priority encoder **426** itself may be omitted, as described above. In alternative embodiments, if it is known that the CMN0 # are to be cleared after assertion, then the multiplexer **428** may be replaced with OR gates (i.e., OR'ing all CMN0 # signals together to produce the output of FIG. 4B) and the radio button circuit **420** may be omitted.

(38) While some of the embodiments described above use the MIPI RFFE serial bus as an example of a serial bus, the invention is not limited to that particular serial bus. As should be clear from the entirety of this disclosure, the example given above of the MIPI RFFE bus is but one of many possible buses (at various clock rates) that may advantageously utilize the inventive concepts of this disclosure. Accordingly, embodiments of the invention may be used with other serial buses, such as the Serial Peripheral Interface (SPI), I2C, I2S, SMBus, and/or PMBus. Further, embodiments of the invention may also be used with applications using parallel bus standards in which a clock signal is associated with parallel address and data busses (e.g., AMBA), or even a combination of serial and parallel busses.

Circuit Embodiments

(39) Circuits and devices in accordance with the present invention may be used alone or in combination with other components, circuits, and devices. Embodiments of the present invention may be fabricated as integrated circuits (ICs), which may be encased in IC packages and/or in modules for ease of handling, manufacture, and/or improved performance. In particular, IC embodiments of this invention are often used in modules in which one or more of such ICs are combined with other circuit components or blocks (e.g., filters, amplifiers, passive components, and possibly additional ICs) into one package. The ICs and/or modules are then typically combined with other components, often on a printed circuit board, to form part of an end product such as a cellular telephone, laptop computer, or electronic tablet, or to form a higher-level module which may be used in a wide variety of products, such as vehicles, test equipment, medical devices, etc. Through various configurations of modules and assemblies, such ICs typically enable a mode of communication, often wireless communication.

(40) As one example of further integration of embodiments of the present invention with other components, FIG. 5 is a top plan view of a substrate **500** that may be, for example, a printed circuit board or chip module substrate (e.g., a thin-film tile). In the illustrated example, the substrate **500** includes multiple ICs **502a-502d** having terminal pads **504** which would be interconnected by conductive vias and/or traces on and/or within the substrate **500** or on the opposite (back) surface of the substrate **500** (to avoid clutter, the surface conductive traces are not shown and not all terminal pads are labelled). The ICs **502a-502d** may embody, for example, signal switches, active filters, amplifiers (including one or more LNAs), and other circuitry. Some or all of the ICs **502a-502d** may include circuits coupled to and controlled by one or more MIPI RFFE serial buses.

(41) The substrate **500** may also include one or more passive devices **506** embedded in, formed on, and/or affixed to the substrate **500**. While shown as generic rectangles, the passive devices **506** may

be, for example, filters, capacitors, inductors, transmission lines, resistors, planar antennae elements, transducers (including, for example, MEMS-based transducers, such as accelerometers, gyroscopes, microphones, pressure sensors, etc.), batteries, etc., interconnected by conductive traces on or in the substrate **500** to other passive devices **506** and/or the individual ICs **502a-502d**.

(42) The front or back surface of the substrate **500** may be used as a location for the formation of other structures. For example, one or more so-called patch antennae may be formed on or affixed to the front or back surface of the substrate **500**; one example of a front-surface patch antenna **508** is shown, coupled to an IC die **502b**, which may include RF front-end circuitry. A patch antenna, and various known variations, is a type of radio antenna with a low profile which can be formed or mounted on a flat surface. Thus, by including one or more patch antennae on the substrate **500**, a complete radio may be created.

System Aspects

(43) Embodiments of the present invention are useful in a wide variety of larger radio frequency (RF) circuits and systems for performing a range of functions. Such functions are useful in a variety of applications, such as radio systems (including cellular radio systems), radar systems (including phased array and automotive radar systems), and test equipment.

(44) Radio system usage includes wireless RF systems (including base stations, relay stations, and hand-held transceivers) that use various technologies and protocols, including various types of orthogonal frequency-division multiplexing (“OFDM”), quadrature amplitude modulation (“QAM”), Code-Division Multiple Access (“CDMA”), Time-Division Multiple Access (“TDMA”), Wide Band Code Division Multiple Access (“W-CDMA”), Global System for Mobile Communications (“GSM”), Long Term Evolution (“LTE”), 5G, and WiFi (e.g., 802.11a, b, g, ac, ax), as well as other radio communication standards and protocols.

(45) As an example of wireless RF system usage, FIG. **6** illustrates an exemplary prior art wireless communication environment **600** comprising different wireless communication systems **602** and **604**, and which may include one or more mobile wireless devices **606**.

(46) A wireless device **606** may be capable of communicating with multiple wireless communication systems **602**, **604** using one or more of the telecommunication protocols noted above. A wireless device **606** also may be capable of communicating with one or more satellites **608**, such as navigation satellites (e.g., GPS) and/or telecommunication satellites. The wireless device **606** may be equipped with multiple antennas, externally and/or internally, for operation on different frequencies and/or to provide diversity against deleterious path effects such as fading and multi-path interference. A wireless device **606** may be a cellular phone, a personal digital assistant (PDA), a wireless-enabled computer or tablet, or some other wireless communication unit or device. A wireless device **606** may also be referred to as a mobile station, user equipment, an access terminal, or some other terminology.

(47) The wireless system **602** may be, for example, a CDMA-based system that includes one or more base station transceivers (BSTs) **610** and at least one switching center (SC) **612**. Each BST **610** provides over-the-air RF communication for wireless devices **606** within its coverage area. The SC **612** couples to one or more BSTs in the wireless system **602** and provides coordination and control for those BSTs.

(48) The wireless system **604** may be, for example, a TDMA-based system that includes one or more transceiver nodes **614** and a network center (NC) **616**. Each transceiver node **614** provides over-the-air RF communication for wireless devices **606** within its coverage area. The NC **616** couples to one or more transceiver nodes **614** in the wireless system **604** and provides coordination and control for those transceiver nodes **614**.

(49) In general, each BST **610** and transceiver node **614** is a fixed station that provides communication coverage for wireless devices **606**, and may also be referred to as base stations or some other terminology. The SC **612** and the NC **616** are network entities that provide coordination and control for the base stations and may also be referred to by other terminologies.

(50) An important aspect of any wireless system, including the systems shown in FIG. 6, is in the details of how the component elements of the system perform. FIG. 7 is a block diagram of a transceiver **700** that might be used in a wireless device, such as a cellular telephone, and which may beneficially incorporate an embodiment of the present invention for improved performance. As illustrated, the transceiver **700** includes a mix of RF analog circuitry for directly conveying and/or transforming signals on an RF signal path, non-RF analog circuitry for operational needs outside of the RF signal path (e.g., for bias voltages and switching signals), and digital circuitry for control and user interface requirements. In this example, a receiver path Rx includes RF Front End, IF Block, Back-End, and Baseband sections (noting that in some implementations, the differentiation between sections may be different).

(51) The receiver path Rx receives over-the-air RF signals through an antenna **702** and a switching unit **704**, which may be implemented with active switching devices (e.g., field effect transistors or FETs), or with passive devices that implement frequency-domain multiplexing, such as a diplexer or duplexer. An RF filter **706** passes desired received RF signals to a low noise amplifier (LNA) **708**, the output of which is combined in a mixer **710** with the output of a first local oscillator **712** to produce an intermediate frequency (IF) signal. The IF signal may be amplified by an IF amplifier **714** and subjected to an IF filter **716** before being applied to a demodulator **718**, which may be coupled to a second local oscillator **720**. The demodulated output of the demodulator **718** is transformed to a digital signal by an analog-to-digital converter **722** and provided to one or more system components **724** (e.g., a video graphics circuit, a sound circuit, memory devices, etc.). The converted digital signal may represent, for example, video or still images, sounds, or symbols, such as text or other characters.

(52) In the illustrated example, a transmitter path Tx includes Baseband, Back-End, IF Block, and RF Front End sections (again, in some implementations, the differentiation between sections may be different). Digital data from one or more system components **724** is transformed to an analog signal by a digital-to-analog converter **726**, the output of which is applied to a modulator **728**, which also may be coupled to the second local oscillator **720**. The modulated output of the modulator **728** may be subjected to an IF filter **730** before being amplified by an IF amplifier **732**. The output of the IF amplifier **732** is then combined in a mixer **734** with the output of the first local oscillator **712** to produce an RF signal. The RF signal may be amplified by a driver **736**, the output of which is applied to a power amplifier (PA) **738**. The amplified RF signal may be coupled to an RF filter **740**, the output of which is coupled to the antenna **702** through the switching unit **704**.

(53) The operation of the transceiver **700** is controlled by a microprocessor **742** in known fashion, which interacts with system control components (e.g., user interfaces, memory/storage devices, application programs, operating system software, power control, etc.). In addition, the transceiver **700** will generally include other circuitry, such as bias circuitry **746** (which may be distributed throughout the transceiver **700** in proximity to transistor devices), electro-static discharge (ESD) protection circuits, testing circuits (not shown), factory programming interfaces (not shown), etc.

(54) In modern transceivers, there are often more than one receiver path Rx and transmitter path Tx, for example, to accommodate multiple frequencies and/or signaling modalities. Further, as should be apparent to one of ordinary skill in the art, some components of the transceiver **700** may be in a positioned in a different order (e.g., filters) or omitted. Other components can be (and usually are) added (e.g., additional filters, impedance matching networks, variable phase shifters/attenuators, power dividers, etc.).

(55) As a person of ordinary skill in the art will understand, a system architecture is beneficially impacted by the current invention in critical ways, including improved design flexibility for systems requiring multiple serial buses.

Methods

(56) Another aspect of the invention includes methods for enabling common control of one or more agent devices by two or more serial buses. For example, FIG. 8 is a process flow chart **800** showing

one method for enabling common control of one or more agent devices by two or more serial buses. The method includes: asserting a write-complete signal when a write operation from one of the two or more serial buses to a common register is completed (Block **802**); and selecting the content of the common register associated with one of the two or more serial buses in response to the assertion of a last write-complete signal (Block **804**).

(57) Additional aspects of the above method may include one or more of the following: wherein at least one write-complete signal is associated with at least two common registers; and/or wherein at least one write-complete signal is associated with a field within one common register.

Fabrication Technologies & Options

(58) The term “MOSFET”, as used in this disclosure, includes any field effect transistor (FET) having an insulated gate whose voltage determines the conductivity of the transistor, and encompasses insulated gates having a metal or metal-like, insulator, and/or semiconductor structure. The terms “metal” or “metal-like” include at least one electrically conductive material (such as aluminum, copper, or other metal, or highly doped polysilicon, graphene, or other electrical conductor), “insulator” includes at least one insulating material (such as silicon oxide or other dielectric material), and “semiconductor” includes at least one semiconductor material.

(59) As used in this disclosure, the term “radio frequency” (RF) refers to a rate of oscillation in the range of about 3 kHz to about 300 GHz. This term also includes the frequencies used in wireless communication systems. An RF frequency may be the frequency of an electromagnetic wave or of an alternating voltage or current in a circuit.

(60) Various embodiments of the invention can be implemented to meet a wide variety of specifications. Unless otherwise noted above, selection of suitable component values is a matter of design choice. Various embodiments of the invention may be implemented in any suitable integrated circuit (IC) technology (including but not limited to MOSFET structures), or in hybrid or discrete circuit forms. Integrated circuit embodiments may be fabricated using any suitable substrates and processes, including but not limited to standard bulk silicon, high-resistivity bulk CMOS, silicon-on-insulator (SOI), and silicon-on-sapphire (SOS). Unless otherwise noted above, embodiments of the invention may be implemented in other transistor technologies such as CMOS, SOI CMOS, bipolar, BiCMOS, LDMOS, BCD, GaAs HBT, GaN HEMT, GaAs pHEMT, and MESFET technologies.

(61) Voltage levels may be adjusted, and/or voltage and/or logic signal polarities reversed, depending on a particular specification and/or implementing technology (e.g., NMOS, PMOS, or CMOS, and enhancement mode or depletion mode transistor devices). Component voltage, current, and power handling capabilities may be adapted as needed, for example, by adjusting device sizes, serially “stacking” components (particularly FETs) to withstand greater voltages, and/or using multiple components in parallel to handle greater currents. Additional circuit components may be added to enhance the capabilities of the disclosed circuits and/or to provide additional functionality without significantly altering the functionality of the disclosed circuits.

Conclusion

(62) A number of embodiments of the invention have been described. It is to be understood that various modifications may be made without departing from the spirit and scope of the invention. For example, some of the steps described above may be order independent, and thus can be performed in an order different from that described. Further, some of the steps described above may be optional. Various activities described with respect to the methods identified above can be executed in repetitive, serial, and/or parallel fashion.

(63) It is to be understood that the foregoing description is intended to illustrate and not to limit the scope of the invention, which is defined by the scope of the following claims, and that other embodiments are within the scope of the claims. In particular, the scope of the invention includes any and all feasible combinations of one or more of the processes, machines, manufactures, or compositions of matter set forth in the claims below. (Note that the parenthetical labels for claim

elements are for ease of referring to such elements, and do not in themselves indicate a particular required ordering or enumeration of elements; further, such labels may be reused in dependent claims as references to additional elements without being regarded as starting a conflicting labeling sequence).

Claims

1. A circuit configured to be coupled to two or more buses, the circuit including: (a) two or more agent devices; (b) at least two decoders, each including: (1) at least two common registers; (2) a bus decoder coupled to the at least two common registers and configured to be coupled to a respective bus of the two or more buses, the bus decoder configured to assert a write-complete signal when a write operation to a corresponding one of the at least two common registers is completed; (c) a first multiplexer having at least two selectable input bus ports, each selectable input bus port coupled to one of the at least two common registers within respective ones of the at least two decoders, and an output port configured to be coupled to at least one of the two or more agent devices; (d) a second multiplexer having at least two selectable input bus ports, each selectable input bus port coupled to one of the at least two common registers within respective ones of the at least two decoders, and an output port configured to be coupled to at least one of the two or more agent devices; and (e) a selection circuit, coupled to two or more of the bus decoders and to the first and second multiplexers, the selection circuit configured to select one of the at least two input bus ports of at least one of the first and second multiplexers in response to the assertion of a last write-complete signal from the coupled bus decoders.
2. The circuit of claim 1, wherein at least one write-complete signal is associated with at least two common registers.
3. The circuit of claim 1, wherein at least one write-complete signal is associated with a field within one common register of the at least two common registers.
4. The circuit of claim 1, wherein the selection circuit includes two flipflops, each coupled to the bus decoder of two of the at least two decoders and to a respective one of the first and second multiplexers, each flipflop configured to select an input bus port of the respective one multiplexer in response to the assertion of a last write-complete signal from one of the two coupled bus decoders.
5. The circuit of claim 4, wherein the output of each flipflop is set in response to the assertion of a write-complete signal from a first coupled bus decoder, and reset in response to the assertion of a write-complete signal from a second coupled bus decoder.
6. The circuit of claim 4, wherein at least one write-complete signal is associated with at least two common registers.
7. The circuit of claim 4, wherein at least one write-complete signal is associated with a field within one common register of the at least two common registers.
8. The circuit of claim 1, wherein the selection circuit includes at least one OR gate, each coupled to the bus decoder of two of the at least two decoders and to a respective one of the first and second multiplexers, each OR gate configured to select an input bus port of the respective one multiplexer in response to the assertion of a last write-complete signal from one of the two bus decoders after the other one of the two bus decoders has been cleared.
9. The circuit of claim 8, wherein at least one write-complete signal is associated with at least two common registers.
10. The circuit of claim 8, wherein at least one write-complete signal is associated with a field within one common register.
11. A circuit that enables common control of one or more agent devices by two or more buses using mutually exclusive arbitration, the circuit including: (a) at least two decoders, each including: (1) at least one common register; (2) a bus decoder coupled to the at least one common register and

configured to be coupled to a respective bus of the two or more buses, the bus decoder configured to assert a write-complete signal when a write operation to a corresponding one of the at least one common register is completed; (b) a multiplexer having at least two selectable input bus ports, each selectable input bus port coupled to one of the at least one common register within respective ones of the at least two decoders, and an output port configured to be coupled to at least one of the one or more agent devices; and (c) a radio button circuit coupled to the bus decoder of the at least two decoders and to the multiplexer, the radio button circuit configured to select an input bus port of the multiplexer in response to the assertion of a last write-complete signal from one of the coupled bus decoders.

12. The circuit of claim 11, wherein the radio button circuit includes at least two flipflops each having an output, the output of each flipflop being set in response to the assertion of a write-complete signal from an associated bus decoder, and reset in response to the assertion of a write-complete signal from any other bus decoder.

13. The circuit of claim 12, further including a priority encoder coupled between the outputs of the flipflops and binary-coded selector inputs of the multiplexer, the priority encoder configured to convert the outputs of the flipflops to a binary-coded value.

14. The circuit of claim 11, wherein at least one write-complete signal is associated with at least two common registers.

15. The circuit of claim 11, wherein at least one write-complete signal is associated with a field within one common register.

16. A circuit configured to be coupled to two or more buses, the circuit including: (a) two or more agent devices; (b) at least two decoders, each including: (1) a common register; (2) a bus decoder coupled to the common register within the decoder and configured to be coupled to a respective bus of the two or more buses, the bus decoder configured to assert a write-complete signal when a write operation to the common register is completed; (c) a multiplexer having at least two selectable input bus ports, each selectable input bus port coupled to the common register within respective ones of the at least two decoders, and an output port configured to be coupled to the at least one of the two or more agent devices; and (d) a selection circuit, coupled to two or more of the bus decoders and to the multiplexer, the selection circuit configured to select one of the at least two input bus ports of the multiplexer in response to the assertion of a last write-complete signal from the coupled bus decoders.

17. The circuit of claim 16, wherein the output of each flipflop is set in response to the assertion of a write-complete signal from a first bus decoder, and reset in response to the assertion of a write-complete signal from a second bus decoder.

18. The circuit of claim 16, wherein at least one write-complete signal is associated with at least two common registers.

19. The circuit of claim 16, wherein at least one write-complete signal is associated with a field within one common register.

20. The circuit of claim 16: (a) wherein the at least selectable two input bus ports of the multiplexer are selectable by binary-coded selector inputs; (b) wherein the selection circuit includes a radio button circuit coupled to the bus decoder of the at least two decoders, the radio button circuit configured to output a set of selection signals in response to the assertion of a last write-complete signal from one of the coupled bus decoders; and (c) further including a priority encoder coupled between the radio button circuit and the binary-coded selector inputs of the multiplexer, the priority encoder configured to convert the set of selection signals output by the radio button circuit to a binary-coded value.
